PassFrame tuition

Perimeter & Area

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 20cm

2. 20cm

3. A — 27cm²

4. 49cm²

5. 28cm

6. 6cm

7. 40cm²

8. 6cm

9. A, B, C, D and E (all of these have a perimeter of exactly 30cm)

10. 36cm

11. 81cm²

12. 30cm²

13. C — both equal 28cm

14. 44cm

15. 8cm

Standard

1. 30cm²

2. 36cm²

3. C — both A (24cm²) and B (24cm²)

4. 24 − 4 = 20cm²

5. 130cm

6. C — 6cm

7. 12000cm²

8. A, B and C (areas 35cm², 32cm² and 36cm² are all more than 30cm²; D=30cm² exactly and E=27cm² are not)

9. 23cm²

10. A — 144m²

11. 36cm

12. 21cm²

13. 150cm² (width 10cm × length 15cm)

14. 96 − 15 = 81m²

15. 24×19 = 456cm²

Standard / Hard

1. 60×35=2100m²; inner = 56×31=1736m²; path = 364m²

2. 216 − 15 = 201m²

3. C — 50cm² (A=36, B=48, D=49)

4. 8cm (area 40cm² ÷ 10 × 2)

5. £240

6. 44m

7. A, C and D (area = 15×4 = 60cm², same as the rectangle, and the same as the triangle: ½×15×8=60)

8. 600 − 64 = 536cm²

9. 13×9 = 117m²

10. Length 10cm, width 7cm

11. 46m (the same as the original rectangle's perimeter)

12. perimeter of 106m×70m = 352m

13. 400m² = 0.04 hectares

14. 16cm (144 ÷ 9)

15. Area = 24+6 = 30m²; cost = £450

Homework — Answers

Easy / Standard

1. 24cm

2. 24cm

3. A — 44cm²

6. 8cm

7. 54cm²

8. 8cm

11. 100cm²

12. 48cm²

13. C — both equal 32cm

4. 64cm²

5. 36cm

9. A, B, C, D and E (all five shapes have a perimeter of exactly 34cm)

10. 38cm

14. 52cm

15. 9cm

Standard

1. 30cm²

2. 55cm²

3. C — both A (30cm²) and B (30cm²)

4. 35 − 6 = 29cm²

5. 140cm

6. C — 8cm

7. 12000cm²

8. A, B, C and D (areas 48cm², 45cm², 42cm² and 42cm² are all more than 40cm²; E=40cm² exactly is not)

9. 30cm²

10. A — 225m²

11. 32cm

12. 40cm²

13. 240cm² (width 7cm × length 16cm)

14. 105 − 12 = 93m²

15. 30×24 = 720cm²

Standard / Advanced

1. Original perimeter = 14cm, so new perimeter = 28cm

2. A — 6cm (36cm² = 6×6)

3. 25cm

4. A, C and D (each of these doubles the area to 48cm²)

5. 64cm² (side 8cm from perimeter 32cm)

6. perim(56,31) = 174m

7. 120cm

8. £378

9. A, B and C (width 7cm, length 14cm, area 98cm²)

10. 52m (the same as the original rectangle's perimeter)

11. 98×53 − 90×45 = 5194 − 4050 = 1144m²

12. 1000m² = 0.1 hectares

13. Length 45cm; perimeter = 100cm

14. Area = 35+6 = 41m²; cost = £656 + £30 = £686

15. 54cm² (area is 3 times as large)